

SCRATCH: THE LOCAL CONFLUENCE LEMMA BEHIND LEMMA *CLIPLOCAL*

Status. Everything in Sections 1–2 is proved. Section 3 states the one class of the local lemma that the computations show to be deterministic; its proof is written as far as it goes and the remaining step is marked. Section 4 is the case tree with counts (all bounded RC graphs of length ≤ 7), and names the sub-cases that are open. Nothing here is in the paper.

1. SETUP

Fix $p \geq 1$. $\mathcal{RC}_{\leq p}$ is the set of bounded RC graphs all of whose crossings lie in rows $1, \dots, p$, of any height $\geq \max(p, \text{maxd})$. For $T \in \mathcal{RC}_{\leq p}$ write $\mathbf{a} = \text{word}(T)$ (rows top to bottom, each row right to left), $w = w_T$, $m = \text{maxd}(w)$. For a position q of $\mathbf{a}$, $\text{rt}(q) = (x, y)$, $x < y$, is the right root: the pair of positions of w whose values are exchanged by the crossing at q , i.e. $\text{rt}(q) = (s_{a_n} \cdots s_{a_{q+1}})(a_q, a_q + 1)$. Distinct positions have distinct roots and $\{\text{rt}(q)\} = \text{Inv}(w)$.

Definition 1.1 (*p*-high cover bumps). A position q with $\text{rt}(q) = (a, b)$ is *p*-high if $a > p$, and is a *cover* if $w t_{ab}$ is a Bruhat cover of w , equivalently $\mathbf{a}$ with the letter at q deleted is reduced (nearly reduced at q). For such q , $\beta_q(T)$ is the Little bump: decrement a_q ; while the word is not reduced, decrement the letter at the unique other position carrying the same root (Little [1, Lemma 4]). Write C_β for the set of positions decremented (the chain) and $q_\beta = q$.

Lemma 1.2. Let $T \in \mathcal{RC}_{\leq p}$, $p \geq 1$, and q a *p*-high cover. Then $\beta_q(T)$ is a valid RC graph in $\mathcal{RC}_{\leq p}$ with the same row lengths as T , $|\beta_q T| = |T|$, and no letter is decremented from 1.

Proof status. Uniqueness of the defect is Little's Lemma 4. Validity of the compatible sequence (letters strictly decreasing within a row, letter $\geq$ row index) and the no-shift statement are verified for length ≤ 7 ; a proof should follow [2, Lemma 3.1] (stack pushes stay within a row) together with the observation that a letter equal to its row index $i \leq p$ has root (x, y) with $x \leq i \leq p$, so cannot lie on a *p*-high chain. *Not written.* $\square$

Definition 1.3 (Canonical bump, D^p). For $T \in \mathcal{RC}_{\leq p}$ with $m = \text{maxd}(w_T) > p$ let $L(T) = \beta_q(T)$ where $\text{rt}(q) = (m, m+1)$ (the unique such q). Then $D^p(T) = L^k(T)$ for the least k with $\text{maxd} \leq p$; this is Proposition *zerocrystal* of the paper.

2. REDUCTION OF CONFLUENCE TO A LOCAL LEMMA

Let $\Phi(T) = \sum_{(i,j) \in T} j$. Every elementary decrement lowers Φ by one, so every *p*-high cover bump lowers Φ by $|C_\beta| \geq 1$.

Proposition 2.1. Suppose that for every $T \in \mathcal{RC}_{\leq p}$ with $\text{maxd}(w_T) > p$ and every *p*-high cover bump $\beta \neq L$ of T there exist $k \geq 0$ and a sequence of *p*-high cover bumps $\gamma_1, \dots, \gamma_r$ ($r \geq 0$) with

$$(1) \quad L^k(\beta T) = \gamma_r \cdots \gamma_1(L(T)).$$

Then $D^p(\beta T) = D^p(T)$ for every *p*-high cover bump β of every $T \in \mathcal{RC}_{\leq p}$; consequently the identity (*clpzero*) of the paper holds and $\text{clip}^p = D^p$.

Proof. Induction on $\Phi(T)$. If $\text{maxd}(w_T) \leq p$ there is no *p*-high inversion and nothing to prove. Otherwise let β be a *p*-high cover bump; if $\beta = L$ then $D^p(\beta T) = D^p(T)$ by definition. Else take k, γ_i as in (1). Since $D^p \circ L = D^p$ wherever L is defined, $D^p(\beta T) = D^p(L^k \beta T)$. As $\Phi(LT) < \Phi(T)$ and each γ_i is a *p*-high cover bump, the induction hypothesis applied successively to $LT, \gamma_1 LT, \dots$ gives $D^p(\gamma_r \cdots \gamma_1 LT) = D^p(LT) = D^p(T)$. Hence $D^p(\beta T) = D^p(T)$.

For (*clpzero*): by Proposition *zerocrystal*, ZY restricted to the top p rows is a composite of *p*-high cover bumps of $Y_{\leq p}$ (each last-descent bump of Y either leaves $Y_{\leq p}$ unchanged or acts on it as one such bump; verified for length ≤ 7 , and this is the content of GAP 2 in the earlier discussion), so $D^p(ZY) = D^p(Y_{\leq p}) = D^p(Y)$. $\square$

Remark 2.2. Computations (length ≤ 7 , 13 931 pairs (T, β)) show (1) always holds with $k \leq 3$ and $r \leq 2$; at the coarser level of a full zeroing step Z (all last-descent bumps until $\maxd$ drops) it holds with Z in place of L , $k \leq 2$, $r \leq 2$. Neither k nor the identity of the γ_i is uniform, see Section 4.

3. A DETERMINISTIC CLASS

Claim 3.1. *Let $T \in \mathcal{RC}_{\leq p}$, $m = \maxd(w_T) > p$, and let $\beta = \beta_q$ with $\text{rt}(q) = (a, m)$, $p < a < m$ (a cover). Then $\maxd(w_{\beta T}) = m$ and*

$$L(\beta T) = \beta'(L(T)),$$

where β' is the bump of $L(T)$ at the position with root $(a, m+1)$. (Verified in all 3540 instances with length ≤ 7 ; in 1921 of them the chains C_β, C_L are disjoint, in 1619 they overlap.)

Proof, as far as written. Write $w = w_T$. Since (a, m) and $(m, m+1)$ are inversions, $w(a) > w(m) > w(m+1)$, and $(m, m+1)$ is the last descent so $w(m+1) < w(m+2) < \dots$.

Permutations. $w_{\beta T} = w t_{am} t_{ca}$ for some $c < a$ with ℓ adding (paper, Lemma *insertworks*); $c < a$ because the chain moves to earlier positions. So $w_{\beta T}(m) = w(a) > w(m+1) = w_{\beta T}(m+1)$ and positions $> m+1$ are untouched: $\maxd(w_{\beta T}) = m$, and L is defined on βT at the crossing of positions $(m, m+1)$, i.e. of the values $w(a), w(m+1)$. On the other side $w_{LT} = w t_{m,m+1} t_{dm}$ for some $d < m$, so $w_{LT}(m+1) = w(m)$ and $w_{LT}(a) = w(a)$ unless $d = a$. The root $(a, m+1)$ of LT carries the values $(w(a), w(m))$ (if $d \neq a$), which is an inversion; it is a cover of w_{LT} because the values strictly between $w(m)$ and $w(a)$ in positions $a < j < m+1$ of w are the same as those blocking (a, m) in w , of which there are none. (The case $d = a$, where L 's chain ends at position a , needs the same check with $w_{LT}(a) = w(m+1)$; then $(a, m+1)$ carries $(w(m+1), w(m))$, which is not an inversion, so in that case β' must be identified differently. In the data this sub-case does not occur with β at (a, m) : whenever $d = a$ the two chains coincide from position a on and the shape is $L(\beta T) = L(T)$; see class (ii) below.)

Words. Both sides have the same permutation: $w_{L\beta T} = w t_{am} t_{ca} t_{m,m+1} t_{em}$ and $w_{\beta' LT} = w t_{m,m+1} t_{dm} t_{a,m+1} t_{fa}$ for suitable c, e, d, f ; equality of these is a statement about the four pipes $w(a), w(m), w(m+1), w(c)$ that has to be checked case by case ($c = d$, $c \neq d$, whether $e = f$). *Not written.* Equality of words then follows from equality of permutations together with equality of rows $2, \dots, p$ (the trim induction of the paper) only if the first-row letters are also matched, so a direct word-level argument is needed: one has to show that the two decrement sets $C_L(\beta T) \cup C_\beta(T)$ and $C_{\beta'}(LT) \cup C_L(T)$ coincide as multisets of positions. In the disjoint sub-case ($C_\beta \cap C_L = \emptyset$, 1921 instances) the data show $C_{\beta'}(LT) = C_\beta(T)$ and $C_L(\beta T) = C_L(T)$, so the claim reduces to: *a Little bump chain is unchanged by decrementing letters at positions outside it whose roots do not share a coordinate with the chain roots.* This is the standard non-interaction lemma and should follow from the description of the defect as the other crossing of the same two pipes. *Not written.* In the overlapping sub-case (1619 instances) the chains share positions and the multiset identity is a genuine rearrangement. *Open.* $\square$

4. CASE TREE

Notation: q_L start of L , C_L its chain; q_β, C_β for β ; $\text{rt}(q_\beta) = (a, b)$. Shapes: $L^k(\beta T) = \gamma(LT)$ abbreviates (1) with $r = 1$, etc. Counts are over all (T, β) , length ≤ 7 , $\beta \neq L$.

class	condition	count	shapes observed
(i)	$a = m$ (so $b > m+1$), $q_L \in C_\beta$	4828	$\beta T = \gamma(LT)$: 3917; $L(\beta T) = \gamma(LT)$: 829; L^2 : 79; L^3 : 3
(ii)	$q_\beta \in C_L$ (then $b = m+1$)	2107	$L(\beta T) = L(T)$: 1934; $L^{1,2,3}(\beta T) = \gamma(LT)$: 173
(iii)	$b = m$, $a < m$	3540	$L(\beta T) = \gamma(LT)$ with γ at $(a, m+1)$: all (Claim 3.1)
(iv)	$a = m$, chains disjoint	1403	$L(\beta T) = \gamma(LT)$: 1333; L^2 : 68; L^3 : 2
(v)	no shared position, chains disjoint	1787	$L(\beta T) = \gamma(LT)$: 1748; $L(\beta T) = \gamma\gamma'(LT)$: 33; L^2 : 6
(vi)	no shared position, chains overlap	174	$L(\beta T) = \gamma(LT)$: 170; L^2 : 4
(vii)	$b = m+1$, $q_\beta \notin C_L$, disjoint	92	$L(\beta T) = \gamma(LT)$: 90; L^2 : 2

In class (i) the shape $\beta T = \gamma(LT)$ (with γ the bump at the transported crossing) is the case where β 's chain passes through q_L and then continues: β “contains” L . In the generic shape $L(\beta T) = \gamma(LT)$ the compensating bump γ is at the crossing of LT that corresponds to q_β ; when $C_\beta \cap C_L = \emptyset$ this is the same word position (its root having been relabeled by L), otherwise it is the crossing of the same pair of pipes; neither rule holds in all cases, so γ is not given by one transport rule.

Open sub-cases (needed for a proof of Proposition 2.1’s hypothesis):

- (1) The exceptional members of classes (i), (ii), (iv)–(vii) requiring $k \geq 2$ or $r = 2$. They are not separated from the generic members by any of the first-order features tested (chain disjointness, containment of starts, shared positions/pipes, whether β lowers maxd, chain lengths, relative order of chain ends): with all fourteen features combined, 24 of 165 feature classes still contain two shapes. The $k \geq 2$ members scale with the length of the descent strip (e.g. $T = \{43 \mid 54\}$ needs $k = 1$, $\{543 \mid 654\}$ needs $k = 2$), so the correct statement is presumably in terms of the full zeroing step rather than single bumps; at that level the exceptional counts are: Z^2 needed in 252 cases, two compensating bumps in 56.
- (2) Class (iii), overlapping chains (1619 instances): the multiset identity of decrement positions.
- (3) Lemma 1.2 (validity, no shift) in full.

Scripts (in `schubmult/_lscripts`): `_diamond.py` (shapes), `_shape_features.py`, `_shape_features2.py` (feature search), `_shape_Z.py` (zeroing-step level), `_clean_class.py` (Claim 3.1), `_gap_checks.py` (Lemma 1.2 data).

REFERENCES

- [1] D. Little, Combinatorial aspects of the Lascoux–Schützenberger tree, *Adv. Math.* 174 (2003).
- [2] S. Billey, A. Holroyd, B. Young, A bijective proof of Macdonald’s reduced word formula, *Algebr. Comb.* 2 (2019).